Beyond the Chatbot: Embedding Context-Aware AI into Your RAD Studio Applications

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Code Partners provides the end-to-end technology and expert guidance to help our customers deliver amazing software.

"...the real value (of AI) is not in the chat...It's the ability for AI to sit inside an application, understand the context, and actually do the work."

Jensen Huang, NVidia

SK-3 Fix for mobile dragging issue

bug/SK-3-fix-mobile-dragging → master OPEN

#5 · Created 5 minutes ago · Last updated 5 minutes ago

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Description

Normal text

Create PR description

This pull request adds a new function `findClosestDragHandleFromEvent` and updates the existing `findClosestDraggableIdFromEvent`.

- Added a new export for `findClosestDragHandleFromEvent` in `use-sensor-marshal/find-closest-draggable-id-from-event.js`.
- Modified the `useTouchSensor` to include handling of finding drag handle and start pending drag.
- Exported the new function `findClosestDragHandle` from use-sensor-marshal module.
- Updated the useSensorMarshal hook to include finding closest drag handle functionality.
- Added a new callback for finding closest drag handle in useSensorMarshal.

Cancel Insert

All are features powered by AI…
All are within your application…
All aware of your application context.

Increasingly, your customers will expect this…

But how?

SmartCore AI Components

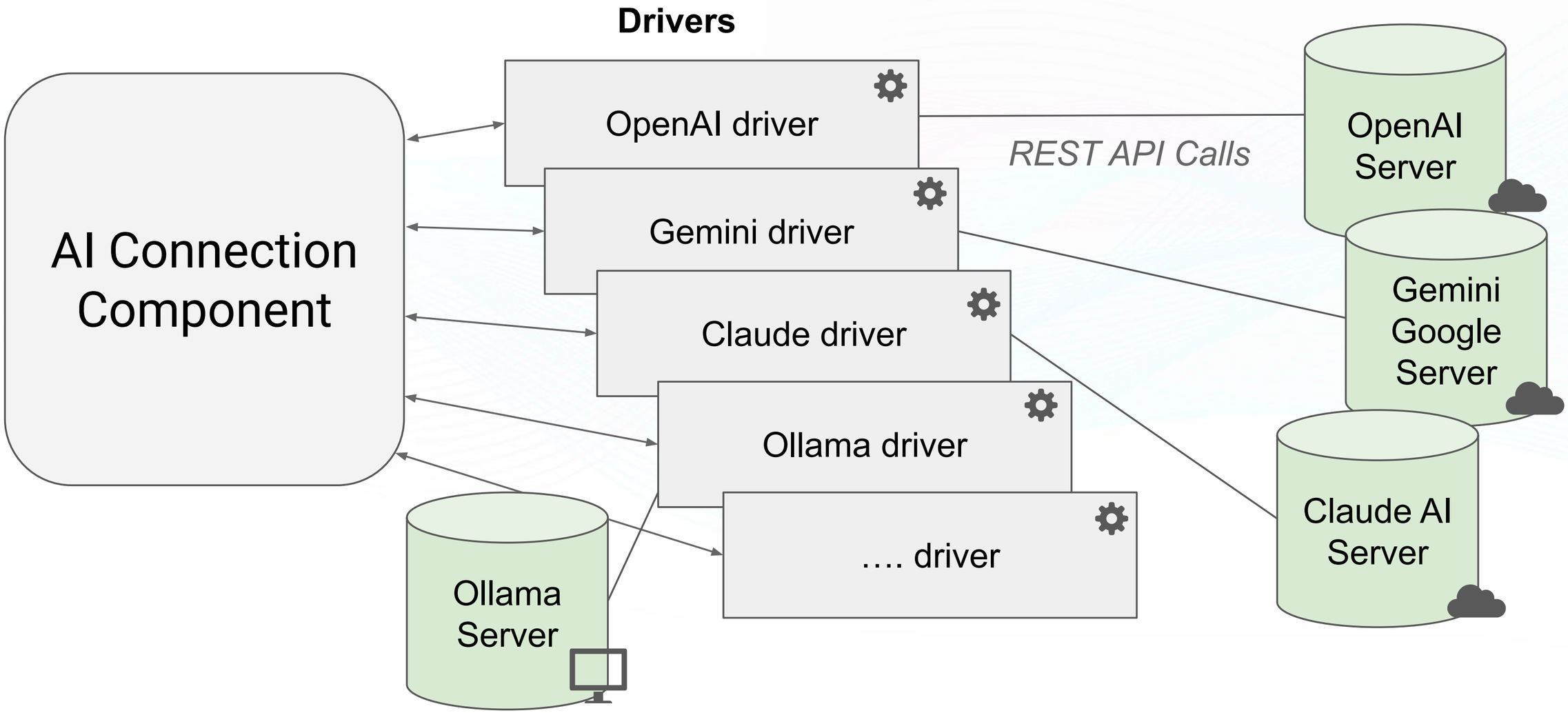
Smart CodeInsight *is Embarcadero's AI for use inside the RAD Studio IDE*
SmartCore AI Component Pack *is Your Gateway to AI inside Your Apps*

A new set of core AI components designed to help customers add AI features to their apps or build brand new AI-focused apps with Delphi or C++Builder – Available in GetIt

Smart AI Components Pack

- **A building block and a foundation for the future**
  - A core architecture our customers and 3rd party vendors can build on and extend to integrate with additional AI engines and UI controls
- **AI connection component (with a role similar to the database connection in FireDAC)**
  - Single AI-engine agnostic component uses specific drivers for AI engines
  - Includes drivers for OpenAI, Claude, Gemini and Ollama, you can add additional drivers for other AI providers
  - Configuration is managed via an easy-to-use AI connection wizard
- **A set of AI request components for different data types**
  - Like text, images, structure data
  - Can be associated with prompts and can be bound directly with UI controls to display the results
- **This component-based model makes it very simple to use the SmartCore AI Components and add them to existing VCL or FMX apps**

Smart AI Components Pack: Drivers Architecture



Smart AI Components Pack: AI Connection Component

- A single component regardless of the backend
- A set of internal drivers to map to different engines
- Expose the drivers' configuration to the connection, with key properties plus some variable ones
- Offer a "connection wizard" to configure a connection and test it out directly
- Possible to add new drivers for additional AI engines (similar to what we offer in the IDE via ToolsAPI for Smart CodeInsight)

Smart AI Components Pack: Frontend

Non-Visual Request Components

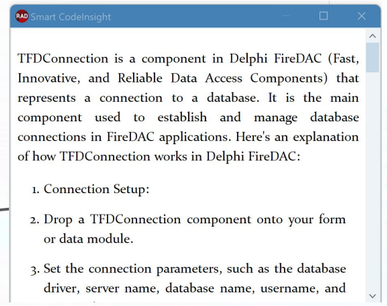
AI Connection Component

Text query

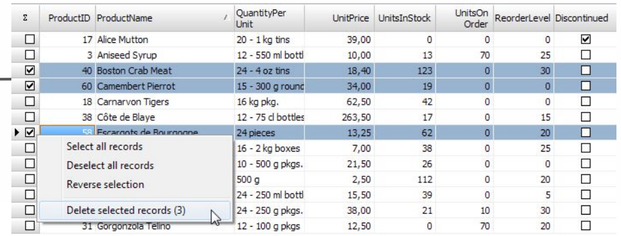
Image request

Structured data request

Drivers for AI Engine



Existing UI components & 3rd party vendors



Smart AI Components Pack: Request Components

- **“Request” components for different data types and processes**
  - Chat / text response supporting plain text and MarkDown
    - Could also include translation requests
  - Image request
  - JSON / structured data request
  - Audio content or stream request
    - Upload audio and receive text
    - Upload text/query, receive audio
    - Upload audio, receive audio
- **Integrates with familiar UI controls: text, image, etc**

Let’s look at some examples

Example Code

- Embarcadero Demos
  - https://github.com/Embarcadero/RADStudio13Demos/tree/main/Object%20Pascal/SmartCoreAI
- My Demos
  - https://github.com/Code-Partners/smartcoreai-demos

Questions

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